Adyansu Beura

Final Year Undergraduate
Department of Electrical Engineering
Indian Institute of Technology Kanpur

Adyansu
Adyansu Beura
adyansub23@iitk.ac.in

Year	Degree/Certificate	Institute	CPI/%
2023 - Present	B.Tech	Indian Institute of Technology Kanpur	**8.6**/10
2023	CBSE(XII)	D.A.V. Public School, Kalinga Nagar	98.6 %
2021	CBSE(X)	D.A.V. Public School, Kalinga Nagar	98.4 %

Scholastic Achievements

- Secured **All India Rank 1034** in the highly competitive **KVPY Aptitude Test (SA Stream) 2021**, conducted by **IISc Bangalore**
- Secured **All India Rank 4** in **National Entrance Screening Test (NEST) 2023,** among more than 44,500 applicants across India
- Awarded the prestigious **National Talent Search Examination (NTSE) Scholarship** officially funded by the Government of India
- Secured **All Odisha Rank 24** in **Junior Mathematics Olympiad 2018**, conducted statewide by the Odisha Mathematical Society

Work Experience

Qualcomm *May'26 – Jul'26*

Interim Engineering Intern

- Analyzed the end-to-end **ASIC implementation flow**, from **RTL synthesis** through **physical design and sign-off**, developing expertise in **timing, routing, and verification methodologies**.
- Executed physical design flows using **Cadence Innovus** across **floorplanning, placement, routing, and CTS**, analyzing critical timing characteristics for large-scale and real-world projects.
- Developed **automation scripts for delay analysis** of **FT nets**, extracting physical-design data and computing **delay metrics** to streamline timing analysis and reduce manual inspection.
- Built an automated **NDR compliance checker for clock networks**, parsing design data to verify the application of **NDR rules** and improve physical-design validation consistency.
- Worked on **backend services and data-handling workflows** supporting **concurrent users** and **large-scale PD data**, gaining experience with **scalable APIs and database interactions**.
- Analyzed **web application scalability and performance**, focusing on **data retrieval, request processing, resource utilization**, and system behavior under increasing workloads.

Self Projects

SpiceLite — C++ Circuit Simulation Engine

- Built **SpiceLite**, a modular terminal-based circuit simulation engine in **C++17**, designing a complete software pipeline from netlist parsing and validation to execution and result generation.
- Designed an **object-oriented** architecture separating circuit components, parsing, solver logic, simulation workflows, and output handling to improve maintainability and extensibility.
- Developed a **SPICE-inspired netlist parser** supporting circuit elements, suffixes, simulation directives, and AC source configurations for converting circuit descriptions into models.
- Established a reproducible development workflow using **CMake, vcpkg, and CTest**, with automated regression tests covering DC, transient, AC, and dependent-source simulation functionality.

Eventrix

- Engineered a comprehensive **full-stack event management platform** using the **MERN stack**, enabling students to discover and register for on university events through a centralized portal.
- Developed secure **RESTful APIs** with **JWT-based authentication** and **Role-Based Access Control**, supporting distinct workflows and personalized dashboards for students and administrators.
- Integrated **payment processing** and **real-time notifications** with a responsive & user friendly interface for seamless registrations.

Key Projects

Large Language Models *Aug'26 – ongoing*

CS781, Prof. Ashutosh Modi

- Implemented and evaluated **Transformer-based** Language Models, exploring **attention mechanisms**, **language modeling**, and modern **LLM architectures** through hands-on assignments.
- Developed practical **LLM pipelines** using **PEFT (LoRA)**, **Retrieval-Augmented Generation (RAG)**, and **quantization**, optimizing model adaptation and inference efficiency.
- Built and experimented with **LLM applications** involving **alignment (RLHF)**, **prompting strategies**, **agentic systems**, and **multimodal LMs**, evaluating performance across diverse tasks.

Predator–Prey Arena *Mar'25 – Apr'25*

EE675 Course Project, Prof. Washim Mondal

- Formulated a **Predator–Prey MDP** with N^4 discrete states and 5 actions, modeling stochastic environment and state transitions.
- Implemented robust **Value Iteration** and **Policy Iteration** using transition-kernel estimation, efficiently handling both dense and sparse matrices across the massive state spaces.
- Trained a neural **REINFORCE policy** integrating gradient clipping and advantage normalization,systematically comparing the performance and convergence behavior of **SGD and Adam**.

Multimodal Sentiment Analysis *Dec'24 – Mar'25*

Google Developer Group, IIT Kanpur

- Developed a **transformer-based multimodal classification system** combining **BERT text embeddings** and **ViT image features** for sentiment, humor, and offense detection in memes.
- Achieved an **82.3% F1-score**, improving performance by **12% over unimodal baselines**, while reducing inference latency to under **500 ms** through model quantization and optimization.
- Evaluated the trained architecture on the IMDB sentiment analysis dataset, achieving **78% classification accuracy** and demonstrating it's ability to generalize effectively across datasets

Graphics and Shaders- 3D Renderer *Dec'24 – Mar'25*

Game Development Club, IIT Kanpur

- Built a real-time **3D graphics engine from scratch in C++** using **OpenGL, GLFW, and GLAD**, implementing the complete graphics rendering pipeline for interactive scene visualization.
- Designed a modular **object-oriented architecture** with reusable abstractions for scene objects, cameras, lights, and rendering components, for organized scene management and resusability.
- Implemented programmable **vertex** and **fragment shaders**, supporting geometric transformations, **Phong lighting**, texture mapping, and dynamic rendering of illuminated environments.

Technical Skills

- **Languages:** C | C++ | Python | LaTeX | Verilog HDL | TCL | Bash
- **Tools and Platform:** Github | Icarus Verilog | Cadence Innovus - PnR | Unity | GTKWave | Microcap | Postman | Hugging Face
- **Libraries & Frameworks:** NumPy | Pandas | Keras| Matplotlib | Scikit-learn | TensorFlow | Seaborn | PyTorch | Langchain

Extra Curricular Activities

- **2nd place**, **Football** representing Pool Nawabs during *Inferno'24*
- **2nd place**, **Institute Cricket Championship (ICC) Tournament**
- Elected as the **Maintenance Secretary** of **Hall 1, IIT Kanpur**

Relevant Courses

(† Ongoing)

Data Structures and Algorithms	Large Language Models†	Introduction to Reinforcement Learning
Probability and Statistics	Fundamentals of Computing	Linear algebra and Differential Equations
Interaction Design in Complex Systems †	Introduction to Economics	Machine Learning for Signal Processing †
Digital Electronics	Control System Analysis	Microelectronics and Power Electronics